

Genetic report for longevity

Attention and ADHD

Polygenic risk of ADHD, catecholamine genes (DAT1, DRD4, COMT, ADRA2A, noradrenaline), response to stimulants and non-stimulants, cognition and sleep

Sérgio F · Report date: 02-10-2026

Hyperactivity and attention deficit disorder (ADHD) is one of the most inherited psychiatric conditions (70–80% in twin studies) and is polygenic: GWAS of 2023 found 27 loci. ADHD is an extreme of continuous traits (attention, impulsivity, restlessness) that exist in the entire population. This report reads two polygenic scores of ADHD, the loci of GWAS, the genes of catecholamines already read in the report Dopamine and Noradrenaline and in the profile (§21, without judgment) and the pharmacogenetics of attention drugs (methylphenidate, atomoxetine, modafinil). Cognition, education and sleep come from the reports Cognitive Capacity and Plasticity and Circadian Rhythm and Sleep. A high score is not a diagnosis: the diagnosis is clinical and requires symptoms since childhood with an impact on life.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

Contents

• Summary	3
• Polygenic risk of ADHD	4
• Catecholamins: dopamine and noradrenaline	6
• Attention drugs	7
• Cognition, sleep and chronotype	8
• Mechanisms of ageing	9
• Practical part	10
• Limitations	11
• Methods and sources	12

Summary

19

variants

10

in the chip

7

imputed
($DR2 \geq 0.8$)

2

not found

3

Scores with
percentile

Overall reading. The genetic predisposition for ADHD is above average and the three scores agree: ADHD (513 thousand variants) at the 94th percentile, ADHD (35,000 variants) in the **97th** and 51st grade of GWAS **79th**. The two polygenic scores come from the same base GWAS and are not independent. In strong loci: PTPRF G/G and ST3GAL3 A/A (2 copies each) against; SORCS3 in favor; FOXP2, DUSP6 and SEMA6D typical. This is not a diagnosis: the score measures continuous traits of attention and impulsivity, and the ADHD genetics overlaps with that of depression (score at the 95th percentile in the Mood report) and that of being considered someone who risks (the 91st percentile). Favourable context: fluid intelligence at the 79th percentile and schooling in the 66 (typical), and morning chronotype in the 86, the opposite direction in the ADHD; against, insomnia in the 89. In catecholamines, without judgment as in §21: DAT1 rs27072 C/C and rs6347 T/T ("increased"), DRD4 –521 C/T, COMT Val/Val, ADRA2A –1291 C/C (without the G allele of best response to methylphenidate), A/A noradrenaline transporter ("reduced"), DBH C/C, SNAP25 G/T, ADGRL3 A/A. In drugs: CYP2D6 without *4 or *10 (atomoxetine in usual dose); CES1 G143E not treated with quality (the pharmacogenetic profile indicates C/C).

Favourable

- **Morning Chronotype (313 loci):** the 86th percentile (Circadian Rhythm and Sleep Report)
- **SORCS3 rs12265655 CT:** SORCS3: neuronal receptor; shared with depression.

Unfavourable

- **ADHD (513 659 variants):** the 94th percentile
- **ADHD (35 445 variants):** the 97th percentile
- **ADHD (loci of GWAS):** the 79th percentile
- **Insomnia (129 loci):** the 89th percentile (Circadian Rhythm and Sleep Report)
- **PTPRF rs549845 GG:** PTPRF: synapse phosphatase; one of the strongest loci of ADHD.
- **ST3GAL3 rs12410444 AA:** ST3GAL3: sialyltransferase; locus of ADHD and intellectual disability.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 10 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Polygenic risk of ADHD

Two ADHD scores (513 thousand and 35 thousand variants, of the same GWAS base) and the GWAS loci score. The strongest loci (PTPRF, ST3GAL3, FOXP2, DUSP6, SORCS3) affect neuronal development and synapse.

Ageing mechanisms: Intercellular communication alteration; Sleep disturbance

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
ADHD (513 659 variants)	Unfavourable	the 94th percentile · z=1.57	Higher score : higher risk. PGS002746 (PGS Catalog; Genome-wide signifiers; Attention-deficit hyperactivity disorder; PGP000358, PMID 35752070). Variants with genotype: 503185 of 513659 (covering of 96.7% coefficients).
ADHD (35 445 variants)	Unfavourable	the 97th percentile · z=1.66	Higher score : higher risk. PGS003753 (PGS Catalog; Genome-wide signifier SNPs; Attention deficit disorder; PGP000473, PMID 36811275). Variants with genotype: 35005 of 35445 (covering of 98.7% coefficients).
ADHD (loci of GWAS)	Unfavourable	the 79th percentile · z=0.73	Higher score : higher risk. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 51 of 54 loci with genotype. Studies: Demontis D (Attention deficit disorder); LaBianca S (Attention deficit hyperactivity disorder); Martin J (Attention deficit disorder); Rovira P (Attention deficiency disorder); Yao X (Attention deficit hyperactivity disorder (MTAG)); van der Laan CM (Attention deficiency disorder). Variants with genotype: 51 of 54 (covering of 94.4% coefficients).
It is considered someone who risks (PGS001049)	No judgement (informative)	the 91st percentile	Information. Source: Personality Report and Psychological Traits.

Favourable

- SORCS3 rs12265655 CT

Unfavourable

- ADHD (513 659 variants) (the 94th percentile)
- ADHD (35 445 variants) (the 97th percentile)
- ADHD (GWAS loci) (the 79th percentile)
- PTPRF rs549845 GG
- ST3GAL3 rs12410444 AA

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SORCS3	rs12265655 10:104984776 · T/C	CT	imputation DR2=0.99	T (1; 77%)	moderate	SORCS3: neuronal receptor; shared with depression. Meaning: GWAS Catalog: Attention deficit hyperactivity disorder (MTAG) (Yao X 2021; p=2e-11); allele T ↑

Unfavourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PTPRF	rs549845 1:43610798 · G/A/C/T	GG	imputation DR2=1.00	G (2; 34%)	moderate	PTPRF: synapse phosphatase; one of the strongest loci of ADHD. Meaning: GWAS Catalog: Attention deficit hyperactivity disorder (Demontis D 2023; p=9e-15); G ↑ allele
ST3GAL3	rs12410444 1:43723048 · A/G	AA	chip	A (2; 71%)	moderate	ST3GAL3: sialyltransferase; locus of ADHD and intellectual disability. Meaning: GWAS Catalog: Attention deficit hyperactivity

Typical / intermediate variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FOXP2	rs9969232 7:114518899 · G/A/C	GA	imputation DR2=0.99	A (1; 61%)	moderate	FOXP2: transcription factor of language; ADHD, risk and posttraumatic stress. Meaning: GWAS Catalog: Attention deficiency disorder (van der Laan CM 2025; p=4e-10); allele A
DUSP6	rs770083 12:89382507 · T/C/G	TC	imputation DR2=0.99	C (1; 44%)	moderate	DUSP6: regulates dopamine signaling (via ERK). Meaning: GWAS Catalog: Attention deficit hyperactivity disorder (van der Laan CM 2025; p=1e-10); allele C
SEMA6D	rs1435746 15:47549801 · G/A	AG	imputation DR2=0.99	G (1; 48%)	moderate	SEMA6D: semaphorin; neuronal development. Meaning: GWAS Catalog: Attention deficit hyperactivity disorder (van der Laan CM 2025; p=9e-09); allele A ↓

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MEF2C	rs3047819 — · —	—	SNP not found in the data	—	moderate	MEF2C: neural transcription factor. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Catecholamins: dopamine and noradrenaline

DAT1 (SLC6A3), DRD4, COMT, ADRA2A, noradrenaline transporter (SLC6A2), DBH, SNAP25 and ADGRL3 (LPHN3). They are the classic candidate genes; the effects are small and they are left without judgment, as in §21. VNTR of DAT1 (10R) and DRD4 (7R) do not read by chip.

Ageing mechanisms: Intercellular communication alteration; Oxidative stress

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (9)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SLC6A3 (DAT1)	rs27072 5:1394407 · C/A/T	CC	chip	—	weak	SLC6A3 3'UTR: "increased" in §21.
SLC6A3 (DAT1)	rs6347 5:1411297 · T/C	TT	chip	—	weak	SLC6A3 exon 9: marker of the haplotype of DAT1 "increased" in §21.
DRD4	rs1800955 11:636784 · T/C/G	CT	chip	—	weak	DRD4 -521C>T: the T allele lowers expression. The 7R allele of VNTR, the classical marker of ADHD, is not read by chip.
COMT	rs4680 22:19963748 · G/A	GG	chip	—	weak	COMT Val158Met Val/Val eliminates prefrontal dopamine faster; working memory benefits more from stimulants and sleep. G/G = Val/Val.
ADRA2A	rs1800544 10:111076745 · G/A/C	CC	imputation DR2=0.99	—	weak	ADRA2A -1291C>G: G allele was associated with the best response of inattention to methylphenidate and is the target of guanfacine. C/C: without the G allele bound to the best response of inattention to methylphenidate.
SLC6A2 (NET)	rs28386840 16:55652906 · T/A/C/G	AA	imputation DR2=0.98	—	weak	SLC6A2 -3081A>T: noradrenaline transporter (target of atomoxetine); "reduced" in §21. A/A: "reduced" in §21 (imputed).
DBH	rs1611115 9:133635393 · T/A/C/G	CC	chip	—	weak	DBH -1021C>T: low T allele to dopamine beta-hydroxylase (conversion of dopamine into noradrenaline); C/C typical in §21.
SNAP25	rs3746544 20:10306436 · G/A/C/T	GT	chip	—	weak	SNAP25 (neurotransmitter release): classic ADHD candidate.
ADGRL3 (LPHN3)	rs6551665 4:61873823 · G/A	AA	chip	—	weak	ADGRL3 (Latrophyllyne 3): was associated with ADHD and the response to stimulants; mixed replication.

Attention drugs

Methylphenidate (eliminated by CES1), atomoxetine (CYP2D6), amphetamines (CYP2D6, in part), modafinil (CYP3A4; induces CYP3A4 and low rapamycin). ADRA2A and ADGRL3 were associated with the response to stimulants.

Ageing mechanisms: Intercellular Communication

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CYP2D6 (*4)	rs3892097 22:42128945 · C/A/G/T	CC	chip	—	strong	CYP2D6*4: null allele; slow metabolizers have several times higher atomoxetine levels. No *4 in §21. C/C: without *4.
CYP2D6 (*10)	rs1065852 22:42130692 · G/A/C	GG	chip	—	moderate	CYP2D6*10: reduced activity. No *10 in §21. G/G: without *10.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CES1	rs71647871 16:55823658 · C/A/T	—	SNP not found in the data (Imputed with DR2=0.54 < 0.8)	A (—; —)	moderate	CES1 G143E: loss of function allele; much higher methylphenidate levels. I do not deal with quality in this chip; the pharmacogenetic profile indicates CC (without G143E).

Cognition, sleep and chronotype

ADHD is associated with the afternoon chronotype, insomnia and lower genetic education; here we join the scores already calculated in other reports to give context.

Ageing mechanisms: Circadian disorders; Sleep disturbance

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Fluid intelligence (PGS001919)	Typical / intermediate	the 79th percentile	"typical" label. Source: Cognitive Capacity and Plasticity report.
Higher education (PGS002685)	Typical / intermediate	the 66th percentile	"typical" label. Source: Cognitive Capacity and Plasticity report.
Morning Chronotype (313 loci)	Favourable	the 86th percentile	ADHD is associated with the afternoon chronotype; here the signal goes in the opposite direction. Source: Circadian Rhythm and Sleep Report.
Insomnia (129 loci)	Unfavourable	the 89th percentile	Sleeping poorly worsens inattention. Source: Circadian Rhythm and Sleep Report.

Favourable

- Morning Chronotype (313 loci) (the 86th percentile)

Unfavourable

- Insomnia (129 loci) (the 89th percentile)

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Sleep disturbance	ADHD (513 659 variants) (the 94th percentile); ADHD (35 445 variants) (the 97th percentile); ADHD (GWAS loci) (the 79th percentile); PTPRF GG; ST3GAL3 AA; Insomnia (129 loci) (the 89th percentile)	SORCS3 CT; Morning Chronotype (313 loci) (the 86th percentile)	FOXP2 GA; DUSP6 TC; SEMA6D AG; Fluid intelligence (PGS001919) (the 79th percentile); Higher education (PGS002685) (the 66th percentile)
Intercellular Communication	ADHD (513 659 variants) (the 94th percentile); ADHD (35 445 variants) (the 97th percentile); ADHD (GWAS loci) (the 79th percentile); PTPRF GG; ST3GAL3 AA	SORCS3 CT	FOXP2 GA; DUSP6 TC; SEMA6D AG
Circadian disorders	Insomnia (129 loci) (the 89th percentile)	Morning Chronotype (313 loci) (the 86th percentile)	Fluid intelligence (PGS001919) (the 79th percentile); Higher education (PGS002685) (the 66th percentile)
Oxidative stress	—	—	—

Practical part

Ideas for attention, crossed with the genetic plan v2 and current medication (isotretinoin, lithium 5 mg, low dose naltrexone 4.5 mg, retatrutide, dapagliflozin, telmisartan, tadalafil, rapamycin). The current stack remains as it is; everything is suggestion (not applied). With a high predisposition, the most helpful is structure, exercise and sleep; drugs only with clinical evaluation.

1. Demanding work in the morning

Genetics: Morning Chronotype p86; ADHD p94–p97

Put the tasks that require more attention in the first hours of the day, with the caffeine of the plan, and leave routine meetings and tasks for the afternoon.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

2. Exercise before difficult tasks

Genetics: ADHD p94–p97; COMT Val/Val

20–30 minutes of aerobic exercise improve attention and inhibitory control during 1–2 hours.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

3. External structure

Genetics: ADHDp94–p97

Time blocks of 25–50 minutes, a unique list of tasks, notifications turned off and the mobile phone outside the desk. They are the strategies with better evidence without drugs.

V2 genetic plan and medication: It's not in the plan. • *suggestion (not applied)*

4. Protecting sleep

Genetics: p89 Insomnia

Sleeping poorly worsens inattention more than any gene; fixed time and melatonin 0.3–1 mg (the dose of the plan).

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

5. Clinical evaluation if there is impact

Genetics: Normal CYP2D6; CES1 not read; ADRA2A C/C

If inattention impairs work or life, an evaluation by a psychiatrist decides. Atomoxetine: usual dose (normal CYP2D6). Methylphenidate: CES1 not treated with quality. Modafinil: induces CYP3A4 and low rapamycin.

V2 genetic plan and medication: Information. • *suggestion (not applied)*

Avoiding or with caution

- Stimulants without prescription (modafinil, amphetamines, "nootropic" imported): modafinil low to rapamycin and all worsen sleep. *suggestion (not applied)*
- Nicotine as a focus stimulant (dependence). *suggestion (not applied)*
- Notifications and screens open during the demanding work. *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. VNTR of DAT1 and DRD4 and copy number variations of CYP2D6 do not read per chip.
- The two ADHD scores come from the same base GWAS and are not independent.
- A high ADHD score is not a diagnosis; the diagnosis is clinical and requires symptoms since childhood with functional impact.
- The genes of the catecholamines are without judgment, as in the profile (§21).
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS002746 (PGS Catalog; Genome-wide signifiers; Attention-deficit hyperactivity disorder; PGP000358, PMID 35752070 · PGS003753 (PGS Catalog; Genome-wide signifier SNPs; Attention deficit disorder; PGP000473, PMID 36811275).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

- Demontis D et al. Genome-wide analyzes of ADHD identify 27 risk loci, refine the genetic architecture and imply severe cognitive domains. Nat Genet 2023.
- Demontis D et al. Discovery of the first genome-wide signifier risk for attention deficit/hyperactivity disorder. Nat Genet 2019.
- Pharaone SV, Larsson H. Genetics of attention deficit disorder. Mol Psychiatry 2019.
- Polanczyk G et al. Association of the adrenergic alpha2A receptor gene with methylphenidate improvement of inattentive dimension in children and adolescents with ADHD. Arch Gen Psychiatry 2007.
- Arcos-Burgos M et al. A common variant of the latrophilin 3 gene, LPHN3, confers susceptibility to ADHD and predicts effectiveness of stimulant medication. Mol Psychiatry 2010.
- Zhu HJ et al. Two CES1 gene mutations lead to dysfunctional carboxylesterase 1 activity in man: clinical significance and molecular basis. Am J Hum Genet 2008.
- Michelson D et al. CYP2D6 and clinical response to atomoxetine in children and adolescents with ADHD. J Am Acad Child Adolescent Psychiatry 2007.
- Coogan AN, McGowan NM. The systematic review of circadian function, chronotype and chronotherapy in attention deficit dysfunction disorder. Atten Hyperact Disorder 2017.
- PGS Catalog: PGS002746 (publication PGP000358, PMID 35752070) and PGS003753 (publication PGP000473, PMID 36811275).
- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

Generated in 02-10-2026 (local time) by the scripts of this report.